Sheharyar

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Research Interests

LLM agents and retrieval-augmented generation; machine learning systems; efficient training and inference on GPUs and clusters. Motivated by two years of building and operating production LLM agent systems.

Education

Sorbonne Université Sept 2025 – Sept 2027 (expected)
MSc Computer Science, DIGIT international programme (taught in English) *Paris, France*

- M1 (2025–26): 12.0/20 on the French scale (10/20 is a pass; 12–14 earns the mention Assez Bien), **ranked 4th of 14**.
- Selected M1 results: System Design and Modeling 93/100 (**1st of 17**); research project at LIP6 86.4/100 (**3rd of 12**); Programming and Networks 84.5/100.
- M1 coursework also included Machine Learning, Cloud Computing, and Data Science, Learning and Applications.
- M2 coursework (2026–27): massively parallel GPU programming (CUDA); parallelism and accelerators for cluster computing (SIMD, multithreading, MPI); high-performance computing; deep learning for visual recognition (PyTorch); methodology for network data analysis; network operations, virtualization and automation.
- M2 ends with a 5–6 month research or industry internship (February to August 2027).

University of Engineering and Technology (UET), Taxila Oct 2020 – Sept 2024
BSc Computer Science, CGPA 3.19/4.00 *Taxila, Pakistan*

- Coursework included artificial intelligence, computer vision, parallel and distributed computing, compiler construction, operating systems, big data analytics and data communication.

Research Experience

Research project, Dioptra team, LIP6 (Sorbonne Université and CNRS) Spring 2026
Supervisor: Prof. Timur Friedman *Paris, France*

- Question: does Caracat, the Rust rewrite of the Caracal high-speed traceroute prober used for Internet topology measurement, behave the same as its C++ predecessor?
- Compared the two engines and validated identical probing behaviour across ICMP, UDP, IPv4 and IPv6.
- Project graded 86.4/100, ranked 3rd of 12.

Industry Experience

Lead Software Engineer, Mavera Oct 2024 – Aug 2026
US startup building an AI market-intelligence platform *Remote*

- Built the platform end to end, then led the developers extending it. Co-built its desktop agent in Rust (Tauri) with tool-calling LLM agents, retrieval-augmented generation and streaming.
- Developed Model Context Protocol (MCP) servers, data pipelines and Python (OpenCV) services that analyse meetings and videos and simulate synthetic audiences and focus groups with LLMs.
- Deployed and operated the services on AWS (Kubernetes on EC2, RDS, S3) with CI/CD as usage grew to 120+ client companies.

Full-Stack Developer, Aveksana Jun 2024 – Oct 2024
Platform for framing research proposals and theses *Remote, Finland*

- Built the platform's new document editor for drafting research proposals; the user base grew from 7K to 10K+.

Full-Stack Developer, Gourmet Gulf LLC Feb 2023 – May 2024
Restaurant group *Remote, UAE*

- Built an operations platform for 16 restaurants and 3 cloud kitchens in 3 countries, with background job queues (Redis, BullMQ).

Freelance Software Engineer (part-time) Feb 2022 – Sep 2025
165+ client projects *Remote*

- Built a RAG chatbot platform (NestJS, Python, Qdrant) and a voice-AI receptionist, among other client systems.

Projects

BirdCLEF+ 2026: acoustic species identification (Kaggle) 2026
Python, scikit-learn, librosa · *code*

- Classified 206 bird species from 35.5K+ recordings with engineered audio features; compared 5 models and reached 0.922 macro ROC-AUC (5-fold cross-validation).

Bina.ai: LLM pipeline for regulator decisions Oct 2026
TypeScript, Mistral API · *code*

- Built the backend for a team of 5: Mistral OCR, embeddings and chat turn data-protection sanction decisions into structured cases, and every quote is checked word for word against the source.

AIWrappers: AI app builder 2025
Next.js, PostgreSQL, Vercel AI SDK · *demo*

- Multi-tenant platform where an agent generates a web app from a plain-language description, with one inference layer across 39 models from 5 providers.

Honors and Awards

- 1st place, Mistral AI LLM × Law Hackathon, Paris (Oct 2026).
- Ranked 1st of 17, System Design and Modeling, Sorbonne Université (2026).

Skills

Programming: Python, C/C++, Rust, TypeScript, SQL; CUDA and MPI (coursework).
Machine learning: PyTorch, scikit-learn, LLM agents, RAG, MCP, Qdrant, LangChain.
Systems: Linux, Docker, Kubernetes, AWS, CI/CD, PostgreSQL, Redis.
Languages: English (IELTS Academic 7.5), Urdu (native), French (beginner).